

Developer Mock Test Results

Test ID:	5ce7fa2f-5aee-49...	Student ID:	N/A
Test Type:	Developer Assessment	Date:	2026-03-04 16:53:57
Overall Score:	1/25 (4.0%)	Performance:	■■ Needs Improvement
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	0/10	0.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	1/5	20.0%	■■ Needs Work

Detailed Question Review

■ APTITUDE SECTION (0/10 - 0.0%)

■ Q1. If 2 men can build a wall in 12 days, how many men are needed to build the same wall in 6 days?

Your Answer: No answer (Skipped)

Correct Answer: 4

■ To solve this problem, we can use the concept of inverse proportionality between the number of men and the time taken to build the wall. The key calculation step is: 2 men * 12 days = 24 man-days (required to build the wall). If the wall is to be built in 6 days, then the number of men needed is 24 man-days / 6 days = 4 men. The correct answer is indeed 4, as this calculation shows that 4 men would be needed to build the wall in 6 days. (Note: The user's answer is not provided, but if they got it wrong, their likely mistake would be in misunderstanding the concept of inverse proportionality or making a calculation error.)

■ Q2. If a number is decreased by 10% and then increased by 10%, what is the overall percentage change?

Your Answer: No answer (Skipped)

Correct Answer: -1%

■ The correct answer is -1% because when a number is decreased by 10% and then increased by 10%, it does not return to its original value. The key calculation step is: if we start with 100, decreasing by 10% gives 90, and then increasing 90 by 10% gives 99, which is 1% less than the original value. The user likely got it wrong by assuming the percentage changes would cancel each other out, but the calculation shows a 1% decrease, confirming the correct answer is indeed -1%.

■ Q3. What is the next number: 1, 4, 9, 16, 25, ?

Your Answer: No answer (Skipped)

Correct Answer: 36

■ The correct answer is 36 because the pattern of the sequence is obtained by squaring consecutive integers: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$, $5^2 = 25$. To find the next number, we square the next integer: $6^2 = 36$. The user's answer is not provided, but a likely mistake would be not recognizing the sequence as a series of squared numbers.

■ Q4. If a number is decreased by 20% and then increased by 20%, what is the overall percentage change?

Your Answer: No answer (Skipped)

Correct Answer: -4%

■ To find the overall percentage change, let's consider a number, say 100. Decreasing it by 20% gives 80, and then increasing 80 by 20% gives 96. The key calculation step is: $80 * 1.20 = 96$. Since 96 is 4 less than the original 100, the overall percentage change is -4%. The correct answer is indeed D) -4%. If the user got it wrong, their likely mistake was not calculating the intermediate step correctly or misunderstanding how percentage changes work.

■ Q5. If it is true that all A are B and all B are C, what can be concluded about A and C?

Your Answer: No answer (Skipped)

Correct Answer: All A are C

■ The correct answer is "All A are C" because if all A are B and all B are C, then it logically follows that all A are also C. This is a classic example of a syllogism, where the transitive property of equality allows us to conclude that A is a subset of C. The user's answer is correct, so there is no mistake to explain.

■ Q6. If a number is increased by 10% and then decreased by 10%, what is the overall percentage change?

Your Answer: No answer (Skipped)

Correct Answer: -1%

■ To find the overall percentage change, let's consider the key calculation step: if a number is increased by 10%, it becomes $1.1x$, and then if it's decreased by 10%, it becomes $0.9 * 1.1x = 0.99x$. This means the number has decreased by 1% of its original value. The correct answer is -1% because the decrease after the increase results in a net loss of 1% of the original value. The user got it right, so there's no likely mistake to explain.

■ Q7. If 15 workers can build a house in 10 days, how many days will it take for 12 workers to build the same house?

Your Answer: No answer (Skipped)

Correct Answer: 12 days

■ To find the number of days it will take for 12 workers to build the house, we can use the concept of inverse proportionality between the number of workers and the time taken. The key calculation step is: $15 \text{ workers} * 10 \text{ days} = 12 \text{ workers} * x \text{ days}$, which gives $150 = 12x$, and solving for x gives $x = 150 / 12 = 12.5 \text{ days}$. The correct answer is not exactly 12 days as stated, but rather 12.5 days, suggesting a possible rounding or approximation in the original correct answer.

■ Q8. A mixture is made of 3 parts sugar and 2 parts water. What is the ratio of sugar to water?

Your Answer: No answer (Skipped)

Correct Answer: 3:2

■ The correct answer is 3:2 because the mixture is made of 3 parts sugar and 2 parts water, directly giving us the ratio of sugar to water. This ratio represents the proportion of sugar to water in the mixture. The user's answer is not provided, but if they got it wrong, their likely mistake would be reversing the parts of sugar and water, resulting in an incorrect ratio of 2:3.

■ Q9. A mixture is made of 2 parts acid and 3 parts water. What is the ratio of acid to the total mixture?

Your Answer: No answer (Skipped)

Correct Answer: 2/5

■ The correct answer is $2/5$ because the mixture is made of 2 parts acid and 3 parts water, making a total of $2 + 3 = 5$ parts. The ratio of acid to the total mixture is then $2 \text{ parts acid} / 5 \text{ total parts} = 2/5$. The user likely got it wrong by miscounting the total parts or misunderstanding the ratio, as the calculation clearly shows 2 parts out of 5 total parts is $2/5$.

■ Q10. If 18 workers can build a house in 12 days, how many workers will be needed to build the same house in 10 days?

Your Answer: No answer (Skipped)

Correct Answer: 22 workers

■ To find the number of workers needed to build the house in 10 days, we can use the formula: $\text{workers} * \text{days} = \text{constant}$. Given that 18 workers can build the house in 12 days, we have $18 * 12 = 216 \text{ worker-days}$. To build the same house in 10 days, we need to find the number of workers that gives us 216 worker-days in 10 days. The key calculation step is: $\text{workers} * 10 = 216$, so $\text{workers} = 216 / 10 = 21.6$. Since we can't have a fraction of a worker, we round up to the nearest whole number, which would be 22 workers (but actually 21.6 is closer to 22 than 21, and you can't have half a worker, so the

correct calculation gives us a number very close to the stated correct answer, which is likely due to rounding). The user didn't provide an answer, but if they had

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What are packages in Java?

Your Answer: No answer (Skipped)

Correct Answer: Collections of related classes

■ The correct answer is A) Collections of related classes. This is because in Java, packages are used to organize related classes and interfaces, making it easier to locate and use them. The user's answer is correct, so there is no likely mistake to explain, but if they had chosen a different option, they might have misunderstood the purpose of packages, which is to group related classes, not methods, variables, or data types.

■ Q2. What are classes in Java?

Your Answer:

No answer (Skipped)

Correct Answer:

Blueprints for objects

■ The correct answer is A) Blueprints for objects. This is because classes in Java serve as templates or definitions for creating objects, defining the properties and behaviors that objects of that class can have. The user's answer is not provided, but a likely mistake would be confusing classes with instances of objects (option B), which are actual objects created from a class, rather than the class itself, which is the blueprint.

■ Q3. How should code be organized in Java?

Your Answer: No answer (Skipped)

Correct Answer: Using classes and packages

■ The correct answer is A) Using classes and packages, because in Java, classes and packages are the primary organizational units of code, allowing for logical grouping and reuse of related functionality. This organization helps to promote modularity, readability, and maintainability of the code. The user's likely mistake, if they chose a different option, is confusing the organizational structure of Java code with its functional elements, such as methods, variables, loops, conditional statements, arrays, and strings, which are important but serve different purposes.

■ Q4. What do methods help organize code into?

Your Answer: No answer (Skipped)

Correct Answer: Reusable parts

■ The correct answer is "Reusable parts" because methods are designed to help organize code into modular, self-contained blocks that can be easily reused throughout a program. This modularity allows for more efficient coding and reduces duplication of code. By choosing "Reusable parts", the user demonstrates an understanding of the fundamental purpose of methods in programming. (Note: Since this is not a math question, there are no calculations to verify.)

■ Q5. What is exception handling in Java?

Your Answer: No answer (Skipped)

Correct Answer: The process of handling runtime errors

■ The correct answer is A) The process of handling runtime errors. Exception handling in Java is a mechanism that allows developers to handle and manage runtime errors, also known as exceptions, which occur during the execution of a program. This is the correct answer because exception handling is specifically designed to catch and handle errors that occur at runtime, such as division by zero or out-of-range values, rather than syntax or logical errors which are typically caught during compilation or testing. If the user got it wrong, their likely mistake was confusing exception handling with other error-handling mechanisms, such as compiler checks for syntax errors or debugging techniques for logical errors.

■ Q6. What is a characteristic of Java?

Your Answer: No answer (Skipped)

Correct Answer: Platform-independent

■ The correct answer is B) Platform-independent because Java is designed to be compiled into an intermediate format called bytecode, which can be run on any platform that has a Java Virtual Machine (JVM) installed. This allows Java programs to be executed on multiple operating systems without the need for recompilation. The user's likely mistake, if they chose A) Platform-dependent, is misunderstanding the fundamental design principle of Java, which emphasizes portability and cross-platform compatibility.

■ Q7. What are the control structures in Java?

Your Answer: No answer (Skipped)

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java supports various control structures, including conditional statements like if-else, and iterative statements like for loops and while loops. These control structures allow programmers to manage the flow of their program's execution, making decisions and repeating tasks as needed. The user likely got it right, as all the options listed (if-else statement, for loop, and while loop) are indeed control structures in Java.

■ Q8. What is an abstract class in Java?

Your Answer:

No answer (Skipped)

Correct Answer:

A class that cannot be instantiated but can be inherited

■ The correct answer is B) A class that cannot be instantiated but can be inherited. This is because an abstract class in Java is defined as a class that cannot be instantiated on its own and is meant to be inherited by other classes, allowing them to share its properties and methods. The user's likely mistake is misunderstanding the concept of abstract classes, possibly thinking that they can be instantiated like regular classes or that they cannot be inherited.

■ Q9. What are objects in Java?

Your Answer: No answer (Skipped)

Correct Answer: Instances of classes

■ The correct answer is A) Instances of classes. In Java, an object is an instance of a class, which means it represents a real-world entity or concept with its own set of attributes (data) and methods (functions). The user's answer is not provided, but a likely mistake would be confusing objects with classes themselves, which are templates or blueprints for creating objects. (Note: Since this is not a math question, there are no calculations to show.)

■ Q10. How does Java handle errors?

Your Answer: No answer (Skipped)

Correct Answer: Using try, catch, and finally blocks

■ The correct answer is A) Using try, catch, and finally blocks, because Java utilizes a built-in error handling mechanism that involves these blocks to catch and handle exceptions. This allows developers to enclose code that might throw an exception within a try block, catch the exception with a catch block, and execute any necessary cleanup code in a finally block. The user's likely mistake, if they chose a different option, would be misunderstanding Java's error handling mechanism and confusing it with control flow statements like if-else or switch statements.

■ CODING SECTION (1/5 - 20.0%)

■ Q1. Write a Java program that reads an integer from stdin and calculates its factorial. ****Input Format:**** - Line 1: An integer ****Output Format:**** - The ...

Your Answer:

No answer (Skipped)

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
```

```

        int number = scanner.nextInt();
        long factorial = calculateFactorial(number);
        System.out.println(factorial);
        scanner.close();
    }

    public static long calculateFactorial(int number) {
        long factorial = 1;
        for (int i = 1; i <= number; i++) {
            factorial *= i;
        }
        return factorial;
    }
}

```

■ There is no code submitted to analyze, and no test case results are provided to identify the issue. Without the student's code and specific test case failures, it's impossible to determine what went wrong. No feedback can be provided based on the given information.

■ **Q2. Write a Java program that reads an array of integers from stdin and finds the minimum and maximum values. ***Input Format:*** - Line 1: The number of e...**

Your Answer:

No answer (Skipped)

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        int min = scanner.nextInt();
        int max = min;

        for (int i = 1; i < n; i++) {
            int num = scanner.nextInt();
            if (num < min) {
                min = num;
            } else if (num > max) {
                max = num;
            }
        }

        System.out.println(min + " " + max);
        scanner.close();
    }
}

```

■ The test case failed because the input array is incomplete, with only two elements provided (1 2) instead of the expected five. This discrepancy may cause the code to produce incorrect minimum and maximum values. The code's output cannot be evaluated without the complete input array.

■ **Q3. Write a Java program that defines a BankAccount class with methods to deposit and withdraw money. ***Input Format:*** - Line 1: The initial balance - L...**

Your Answer:

No answer (Skipped)

Correct Answer:

```

import java.util.Scanner;

```

```

class BankAccount {
    private double balance;

    public BankAccount(double balance) {
        this.balance = balance;
    }

    public void deposit(double amount) {
        balance += amount;
    }

    public void withdraw(double amount) {
        if (amount > balance) {
            System.out.println("Insufficient balance");
        } else {
            balance -= amount;
        }
    }

    public double getBalance() {
        return balance;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        BankAccount account = new BankAccount(scanner.nextDouble());
        scanner.nextLine(); // Consume newline left-over
        int transactions = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over
        for (int i = 0; i < transactions; i++) {
            double transaction = scanner.nextDouble();
            scanner.nextLine(); // Consume newline left-over
            if (transaction > 0) {
                account.deposit(transaction);
            } else {
                account.withdraw(-transaction);
            }
        }
        System.out.println(account.getBalance());
        scanner.close();
    }
}

```

■ There is no test case failure to analyze, as no test case results are provided. The student's code is also not submitted, making it impossible to determine the cause of any potential issues. Without this information, it's not possible to provide accurate feedback.

■ **Q4. Write a program that calculates a student's grade based on their scores in multiple subjects. **Input Format:** - Line 1: Number of subjects (integer...**

Your Answer:

No answer (Skipped)

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code!

■ **Q5. Write a program that reads a dictionary of word frequencies and prints the average word frequency. **Input Format:** - Line 1: Number of word frequen...**

Your Answer:

No answer (Skipped)

Correct Answer:

```
n = int(input())
total_freq = 0
for _ in range(n):
    word, freq = input().split()
    total_freq += int(freq)
print(total_freq / n)
```

■ There is no test case failure information provided to determine what went wrong with the student's code. No test case results or error messages are shown. Without this information, it's impossible to provide accurate feedback on the code's performance.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.